

Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

This program will conduct spectroscopy on a supernova using the “B&C” Spectrograph on the Bok 90” telescope at Kitt Peak, Arizona. Due to the transient nature of supernovae and the need for them to be observed within around thirty to forty days of peak brightness, the exact scientific goal of the spectra project will not be known until the observations are conducted in Mid-October 2026. However, we can anticipate that the spectroscopy might allow us to categorize the supernova if it is caught early enough, which would be a significant scientific result. We would characterize the supernova using the SNID-Sage software, which allows us to identify defining features of supernova types, such as the presence of Hydrogen and/or Helium lines. In general, we would be fitting our collected spectrum with the program, to detect elements and/or the radial velocity of our target via Doppler shift. Due to the unpredictability of supernovae and their relatively ephemeral brightness, this proposal offers a sample outline and representative targets, but the ultimate target and science goal will be determined by availability at the time of observation.

Telescope Requested (0.5 points): BOK 90”

Instrument (0.5 points): “B&C” Spectrograph

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

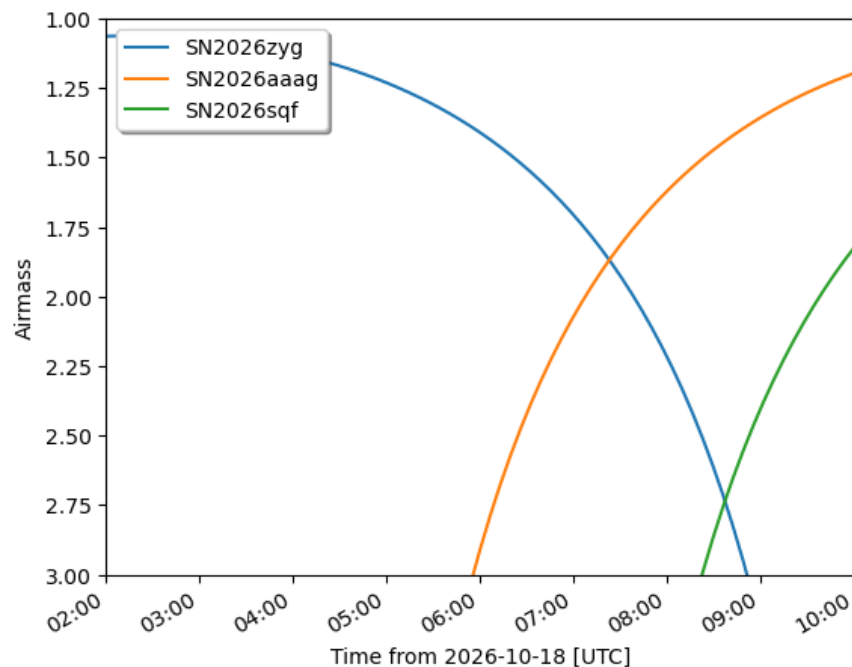
Assuming our time of observation is Oct. 17-19, around those days, the sun will have an RA of around 14 hours and the moon’s RA will be between 18 and 20. This is not ideal for our observations as this means the moon will be reaching its peak right around sunset, however via the astropixels website provided, we expect the moon brightness on these dates to be from 44% to 63% illuminated as the weekend progresses. Thus, it would be more ideal to conduct our observations on October 17th if possible, as this is when the moon will be least bright and will have the least interference with our dim supernova observations. Because our targets will be so dim, the moon will need to be decently far from our supernovae targets so that we can maximize our SNR. The moon will set between 11:00PM and 1:00AM on all of these dates, so it may be advantageous to observe a supernova with an RA around

22-24 or 0-3 hours as these will be crossing the meridian while both the sun and moon are below the horizon.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example “Object can be observed Feb-March.”

Object Name	Right Ascension	Declination	V magnitude at peak	Peak Date	Object Type	Visibility
SN 2026sqf	140.673888	53.37	12.9	7/28	SN Ia	Ursa Major
SN 2026aaag	102.05	51.23555556	18.08	8/29	SN type SN II	Lynx
<u>SN 2026zyg</u>	308.8761	51.5900	18.049	8/28	SN II	Cygnus



The plot shows the airmass for several of the targets outlined in the table for the night of October 17th from 7:00PM to 3:00AM Tucson time (sorry the x-axis on the plot

is in UTC). In October, astronomical twilight ends around 8:00PM so that is when our actual observations will start. The target in orange would be a better choice for a target in our case because it reaches peak later in the night when the moon will be set, and it gets very high in the sky which improves viewing with less airmass. The blue supernova is also a good target for earlier in the night, we would just have to worry more about the separation from the moon.

The targets listed here are good to observe now based on magnitude, but by the time we observe in October these supernovae will likely be too dim to observe. After its peak brightness, a supernova will have about 40 days of dimming until it is too dim to detect. Most of our listed targets, even though discovered relatively recently, are right within this range. More than likely, we will have to adjust the targets closer to our observing date.

Experimental Design & Technical Description (4.5 points total)

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total) Add your name to the file name (_LastNameFirstName.docx) then upload to D2L

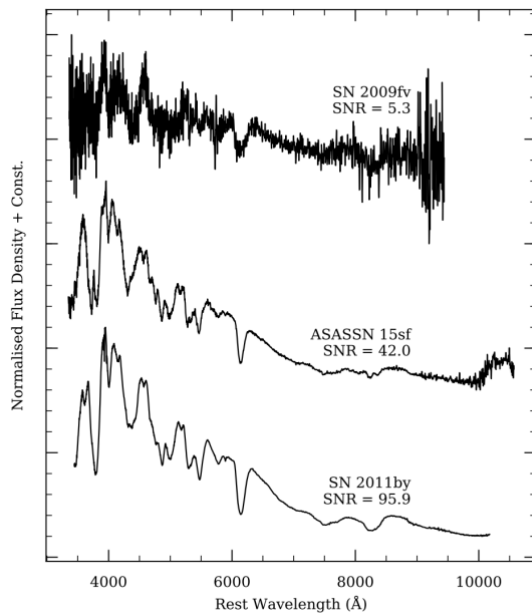
How much time is required for your program and what signal-to-noise is necessary for your observations? **(2 points)**

In Stahl et. al (2020), a paper that observed Type Ia supernovae at varying signal-to-noise ratios and the visibility of distinct spectral features at each, they used a 3-meter telescope pointing at a target with varying airmasses and exposure times. One object, [SN 2009D](#), was pointed at for 1200s in an air mass of 1.86 to get a SNR of 8.3 (at the wavelengths 3390-10700Å). At discovery on January 2nd, SN 2009D had a 16.5 magnitude. This paper's spectra were taken 5 days before its peak value, also in January, so we will assume 16.5 magnitude is the magnitude they saw. To get the same SNR we can find exposure with following relation:

$$t_{BOK\ exp} \sim \left(\frac{D_{KAST}}{D_{BOK}}\right)^2 t_{KAST\ exp}$$

BOK, a smaller 2.3 meter telescope, with a similarly bright source (say 16.5 magnitude) and pointing through similar airmass (we would try to point at 1.5 maximum) would need to be on a target for around, at least, 2040s total exposure to get the same SNR of 8.3. This exposure time would increase as the magnitude dimmed by the $10^{\frac{m_{BOK}-m_{KAST}}{-2.5}}$ factor. The way these two spectrographs function over their wavelength range also differs, so this is a very rough estimate making broad assumptions. In order to detect distinct

spectral features like Hydrogen and Helium within the supernova, we will want a minimum signal to noise ratio of 5, ideally higher.



How will these observations contribute toward the accomplishment of the goals outlined in the abstract? **(2 points)**

The spectrum of the supernova will allow us to detect elements present in the object, which is used to categorize the supernova type. For example, if there are no Hydrogen features in the spectrum, we will know that it is a Type I supernova, and the ionized Silicon (Si II) feature is an indicator of a type Ia supernova specifically. We could do this analysis to either characterize a new supernova (if we are so lucky) or to confirm another person's characterization. Again, we would use the SNID-Sage program to fit the spectrum and identify key features that can inform us of the type and age of the object. Computations like optical depth can give us an idea of the abundances of certain elements within the supernova, and through the doppler shift of spectral lines we can infer the radial velocity.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. **(0.5 points)**

All of us in this group were interested in a supernova that was discovered by BOK earlier this year by Professor Nathan Smith's group. It was a broad-lined, type Ic supernova. So if we can't get any supernova targets we were hoping we could try and get their data or any other past SN they have looked at. In general, if we cannot gather our own supernova spectra we would see if teams at the UofA have data we could use.